

Local Document Intelligence - Architecture Verification Brief

1. Overview

This document serves as an example verification artifact for the Local-First RAG pipeline. It demonstrates that the document extraction engine accurately reads multi-page and single-page PDF files, computes vector embeddings, and links retrieved snippets back to exact page numbers.

2. Zero-Exfiltration Security Guarantees

- All vector indexing operations are isolated to local storage.
- Token generation is processed strictly on local GPU/CPU hardware.
- No network telemetry or external API requests are performed.

3. Quantization & VRAM Benchmarks

Using 4-bit Quantized Llama-3.2 3B and nomic-embed-text, the complete system uses approximately 3.2GB of VRAM, leaving abundant overhead on standard 8GB VRAM consumer GPUs.